Coordination II: Collective Intelligence

1. Sharing the work, the knowledge and the tools

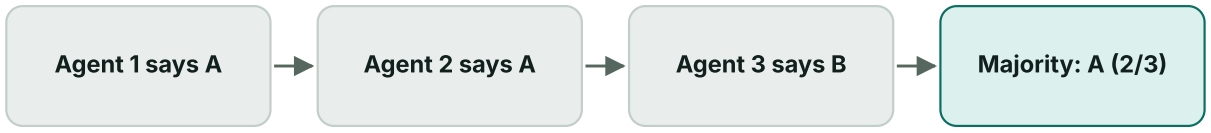
- **Multi-Agent Planning**: agents (or a planner) decompose a goal, assign sub-tasks and sequence dependencies before executing. Watch for re-planning when reality diverges
- **Knowledge Sharing**: a shared store of solved problems agents contribute to and draw from, so the team stops re-solving the same things
- **Tool Routing**: a layer that sends each tool request to the right provider or agent when there are many of both
- Common thread: these externalise what one agent keeps in its head (the plan, the knowledge, the tool map) into shared structures

2. Consensus and negotiation

- **Consensus**: cooperative agents debate, critique and vote (majority, weighted, structured debate) to converge on one answer, e.g. analysts converging on a forecast
- **Negotiation**: self-interested agents exchange offers and counter-offers until they reach a deal over a shared resource. Needs protocols, deadlines and fallback rules

> ⚠ Voting only reduces error when agents' mistakes are **independent**. Three copies of the same model with the same blind spot just repeat the mistake.

✎ DIAGRAM



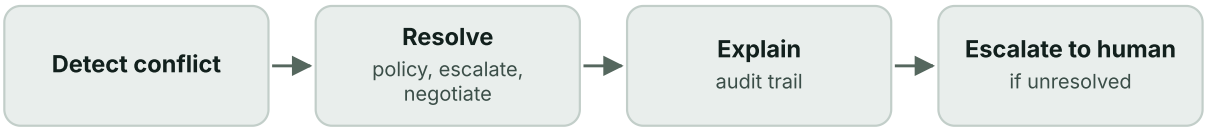
Consensus helps only if the agents' errors are independent.

3. Keeping order under contention

- **Resource Allocation**: market-based, priority-based or optimisation-driven assignment of limited resources (e.g. robots across factory tasks)
- **Conflict Resolution**: when agents reach incompatible decisions: escalate to a supervisor, apply a policy, negotiate, or use game theory. Essentials: **detect** the conflict, **explain** the resolution (audit trail), and keep a **human escalation path**

- **Formation Control**: local rules that keep physical agents in a coordinated shape (e.g. a drone swarm covering a field)

✎ DIAGRAM



Detect, explain, escalate: the governance instinct that recurs across the book.

SAY IT

"Consensus is cooperative agents converging on the best answer; negotiation is competing agents converging on a deal. Either way I'd log the resolution and keep a human escalation path."